

LEARNING OUTCOMES!

Candidates should:

- understand the meaning of the terms raw materials, refined, processed, manufactured and valueadded
- understand the meaning of the terms infrastructure and services
- be able to define the terms primary industry, secondary industry, and tertiary industry
- Processing and manufacturing industries to be studied: cement, cotton (from ginning to clothing), sugar refining, crafts, fertiliser, iron and steel, brick, oil refining, sports goods, surgical instruments. Tourism.
- understand the definitions used in Pakistan to distinguish between large-scale industry, small-scale industry and cottage industry
- state the main products of the listed industries and whether they are destined for the domestic market and/or for the export market
- state the main locations of the listed industries and explain the factors influencing location and development - capital, site, sources of raw materials, power, water, labour, communications, government policy and other means
- understand the differences between the formal sector and informal sector of industry
- understand the range of services provided by the informal sector, and their advantages and disadvantages to the development of Pakistan
- understand the importance of both the formal and informal sectors, and evaluate the contributions of both sectors to the development of the listed industries
- understand sources of capital and labour
- state and explain how the governing authorities promote industrial growth
- name examples of export processing zones and other industrial estates, explain the reasons for their development and describe their characteristics
- assess the feasibility of using global communications to enhance employment opportunities in service industries, e.g. call centres
- state and describe briefly, with an example of each, some of the natural and cultural attractions of Pakistan that are, or could be made available to tourists

- assess the feasibility of developing tourism as a means of increasing employment, development, gross national product (GNP) and gross domestic product (GDP).

DEFINITIONS

Raw Material

- ✓ The basic material from which a product is made.

Refined

- ✓ Impurities or unwanted elements having been removed by processing.

Processed

- ✓ Perform a series of mechanical or chemical operations on (something) in order to change or preserve it.

Manufactured

- ✓ Produced on a large scale using machinery.

Waste Material

- ✓ Unused / unwanted by-product

Value Added

- ✓ The addition of features to a basic line or model for which the buyer is prepared to pay extra.

Infrastructure

- ✓ The basic physical and organizational structures and facilities (e.g. buildings, roads, power supplies) needed for the operation of a society or enterprise.

Services

- ✓ The action of helping or doing work for someone.

INDUSTRY

- ✓ A manufacturing unit is known as industry.

PRIMARY INDUSTRY

- ✓ Primary industry is defined as an industry that is concerned with extracting the natural resources on the earth so that they can be converted into consumable products. It can be the mining of natural resources like wood, iron, coal, minerals, agriculture industry, and even fisheries.

SECONDARY INDUSTRY

- ✓ A secondary industry is an industry that takes raw materials as input and creates finished products as output.

TERTIARY INDUSTRY

- ✓ The part of an industry concerned with the provision of services.

CLASSIFICATION OF INDUSTRIES

Large Scale

- ✓ These are industries which employ more than 300 workers (large number of workers) and their fixed assets exceed 100 million rupees (high capital input).
- ✓ They use many machines and the products are produced in large amounts.
- ✓ Land requirements are big too

Advantages

- ✓ Employment opportunities for many thousands can be generated, which increases their standard of living and contributes more to the GDP
- ✓ Things like steel can be produced locally in large quantities, which is cheaper than imported steel for locals. This can help to fuel industrialization (building factories, dam, railways etc)
- ✓ Exports can be increased, which increases foreign exchange reserves. They are used to pay off foreign loans, if a country defaults in these payments it could risk a trade embargo from the countries it owes money to

- ✓ Foreign dependence on certain goods like iron and steel can be dramatically reduced; allowing for greater independence in foreign affairs. This is because your country would not be influenced by the demands of another country

Disadvantages

- ✓ Air pollution may occur due to exhaust from chimneys and also traffic congestion near the factory may also be of concern. Traffic may consist of trucks bringing in raw materials or taking out products and bringing them to markets
- ✓ Water pollution may also occur if raw untreated waste is dumped. This will mean that the sewage enters the water table and possibly the human food chain and can cause things like cancer
- ✓ Roads may be damaged as they are used more than their capacity with trucks carrying heavy loads most of the time
- ✓ Also, lands may be cleared to make new roads and industrial centres. This will result in deforestation and soil erosion
- ✓ If industries are set up excessively in urban areas then it may cause rural-urban migration (as government will be diverting funds from development of rural areas towards maintenance of urban areas)

Small Scale

- ✓ These are industries which employ 10-50 workers.
- ✓ Their fixed assets don't exceed 10 million rupees.
- ✓ They are labour intensive with limited use of machinery.
- ✓ Some of the raw materials may be brought from nearby cities

Cottage

- ✓ This is a type of industry where 1-5 employees are involved (family members) and fixed assets do not exceed from Rs 1-2 lakh (0.1-0.2 million).
- ✓ They include potter making, furniture, carpet making, embroidery etc.
- ✓ They are labour intensive with little use of machinery.
- ✓ Local skills along with locally available raw materials are utilized

Advantages

- ✓ They provide large scale employment to the relatively poor people, who work in the industrial sector. They can also employ rural artisans, women etc

- ✓ Also, some farmers can increase their income by learning other skills like pottery, which is a form of cottage industry
- ✓ It helps **decrease** regional disparity as the income from these cottage industries is distributed amongst a lot of people
- ✓ These industries have few specific requirements (like large amount of capital etc), thus they can be setup by anyone anywhere.
- ✓ They are labour intensive and that they can accommodate to a large extent illiterate workers. These workers can be easily trained in a matter of few weeks (unlike doctors etc)
- ✓ These industries can produce a significant amount of consumer goods as well as some luxury goods (carpets etc). These goods both compete with large scale industrial manufactured goods. Thus they help reduce the price offered to the final consumer
- ✓ Furthermore, they also help fill the gap between the supply and demand in the market that might exist for a specific good. Imports can thus be reduced
- ✓ Goods that are produced from these industries may also be exported like Pakistani carpets etc; earning foreign exchange
- ✓ Setting up of cottage industries utilizes savings thus they are put into circulation as investments, which benefit many people. Also, it encourages small entrepreneurs to take risks etc thus it develops their ability to invest efficiently
- ✓ Traditional skills can be preserved and passed to the next generation, thus, maintaining cultural identity
- ✓ Eventually these cottage industries help reduce though **not** stop "Rural-Urban Migration". Thus the problems of large, overcrowded cities filled with jobless people can be reduced

Problems

- ✓ Finance management is extremely important for people who set up these cottage industries as their working capital is low. It means that they have less money at their immediate disposal to buy raw materials etc. Financial management requires skill and some financial support from the government
- ✓ The goods produced usually do not meet specific quality requirements thus the first batch of goods may be of excellent quality but the following one may be poor. This can result in sudden fall of demand as customers are lost
- ✓ Also, the marketing of goods is expensive and is a huge burden on the relatively poor cottage industrialist

- ✓ Producers in this sector often purchase goods in small quantities, thus they cannot benefit from the economies of scale, which the big companies enjoy when they buy raw materials in bulk. Thus, the cost of producing a single unit can be high
- ✓ The raw materials available in that specific area might be of poor quality, leaving the cottage industry with a dire choice of either using them or facing bankruptcy, as they are too ill-equipped financially to buy raw materials from other areas
- ✓ These cottage industries are mostly labour intensive and have little or low reliance on expensive machines. Machines used by large scale industries can produce variety of goods in a short amount of time
- ✓ On the other hand, a skilled cottage industry worker takes years to master the art of making a specific object

Explain how government organisations help and promote the development of small-scale industries.

- ✓ Organisations e.g. PSIC (Pakistan Small Industry Corporation), PSIC (Punjab SIC), SMEDA (small and medium enterprise development agency) etc.
- ✓ marketing facilities
- ✓ Trade fairs
- ✓ Education and training
- ✓ cheap loans/loans on easy instalments
- ✓ tax breaks/cheaper raw materials
- ✓ small industrial estates
- ✓ dry ports/better road transport
- ✓ power supply/electrification/gas/water
- ✓ telecommunications
- ✓ aid for mechanization

To what extent can small scale and cottage industries be sustainable? Explain your answer. [6]

Possibilities

- ✓ Can be done in the home (so low set up costs)
- ✓ Local raw materials / re-used waste materials (therefore cheap / less need to borrow money)
- ✓ Simple technology (so less need for electricity / power supply)
- ✓ Small scale causing less damage to environment

- ✓ Traditional skills / does not demand education (people can support themselves when other work not available)
- ✓ Government support / schemes / loans
- ✓ Use of machinery
- ✓ Use of electricity (allowing work after dark)
- ✓ Training / education

Problems

- ✓ Small output / low earnings
- ✓ Low profits (therefore difficult to escape poverty)
- ✓ Need for 'middle man' which reduces profits
- ✓ Poor quality products (so lack of demand / low value)
- ✓ Use of child labour (restricting markets / sales)
- ✓ Lack of education / telecommunications (which limits marketing skills / limit marketing ability)

FACTORS AFFECTING THE LOCATION

Site Requirements

- ✓ Land must be cheap. This will mean that cost of setting up industry will be low and will encourage investors to invest
- ✓ Land must be abundant so if the owners want to expand their existing businesses they can easily do so
- ✓ The land must be well drained so during times of heavy rainfall the factory isn't flooded and production isn't stopped

Transport/Infrastructure

- ✓ Bulky, cheap and quick transport of both raw materials and product is available.
- ✓ This will mean that the per unit cost of making a product and transporting it (to the market) are low.
- ✓ This gives the company a pricing edge over its competitors.
- ✓ This will increase the profits of a company

Market Access

- ✓ A large demand for goods produced is present nearby.
- ✓ Nearness to the market reduces the cost of transport.
- ✓ If a large market is present, then the company can make bigger profits due to economies of scale
- ✓ The demand is stable and not volatile, so it doesn't vary throughout the year. This means that the company will have sales throughout the year and will have money throughout the year to pay its creditors/ contractors

Raw Materials

- ✓ Stable supply of raw material is present; so the company can order supplies if it faces high demand. This will improve the company's image of a reliable source of products and will help improve sales
- ✓ Large amount of cheap raw material is available near to the factory. Nearness reduces chances of accidents and reduces cost of transport. This will mean that per unit cost of product is low, giving the company a pricing edge over its competitors.
- ✓ This will increase the profits of a company

Labour

- ✓ Large pool of skilled and unskilled labour is required, which needs little or no further training. This reduces induction costs and time; leading to smoother operations
- ✓ An educated labour force can easily use machines by reading instructions. Also, they use their skills to suggest improvements to work practices. They have the necessary skills, which they have learnt from polytechnic institutes.
- ✓ Wage costs are low so that per unit cost of making a product is low, giving the company a pricing edge over its competitors.
- ✓ This will increase the profits of a company
- ✓ Skilled labour is required to install and operate machines, do managerial work etc.
- ✓ Unskilled labour is required for driving trucks (transporting raw materials and products) etc

Power

- ✓ Cheap source of power is available, which means that the per unit cost of making a product is low, giving the company a pricing edge over its competitors.
- ✓ This will increase the profits of a company.
- ✓ The sources of power must be reliable so machines aren't damaged due to sudden spike in voltage etc

Capital

- ✓ Cheap (interest free loans are the best) sources of capital are available from either the state or private banks. This will allow the business to invest in fixed assets etc.
- ✓ The timeframe for making payments must be wide, so giving a business some breathing space. Loans must be easy to get with no strict conditions for getting a loan, so people are encouraged to invest in their businesses
- ✓ These assets require a lot of capital to set up, but their benefits are long lasting. They include land, telecommunications, power/gas/water supply (cleaning, cooling, drinking etc) and machinery (cheaper than labour, more accurate, quick and sturdy)

Communication

- ✓ Cheap communication services must be available. These link the customer and the producer, which is essential to the success of a business

Government Policies

- ✓ Industrial Estates
- ✓ Export Processing Zones

Explain how human inputs can improve production.

- ✓ Labour - work machines, carry materials, office work
- ✓ Capital - wages, machines, technology, investment
- ✓ Machinery - faster, better quality, new products
- ✓ Skills - computers, office work, machines
- ✓ Technology - quality, speed, modernisation
- ✓ Transport - faster, larger supply, bigger markets
- ✓ Power - efficiency, speed, quality
- ✓ Water - for cleaning
- ✓ Government policies- tax concessions, training

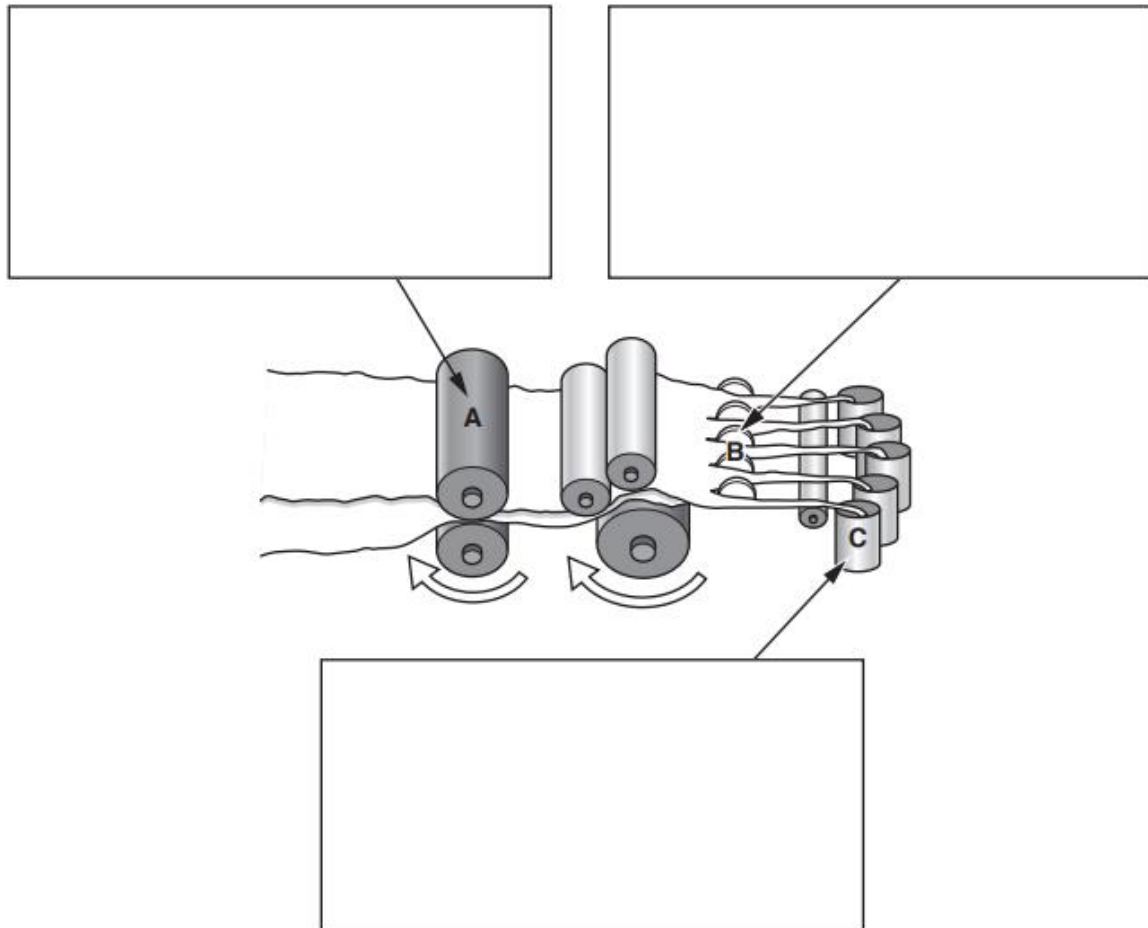
GOVERNMENT WORKS TO ATTRACT INVESTORS

- ✓ Industrial estates Example eg. Sindh Trading Estate Ltd. (SITE)
- ✓ Tax exemptions on imported machinery
- ✓ Less foreign exchange control
- ✓ Tax holidays
- ✓ Simplified procedures
- ✓ Private power stations Example Hub (?Hab) power project
- ✓ Better roads/railways/airports
- ✓ Dry ports for better security
- ✓ Agencies to help investors provide infrastructure e.g. water, electricity, telephone, roads to estates
- ✓ Land available for housing, commercial, social facilities near industrial zones
- ✓ Consistent policies/stable government

COTTON TEXTILE INDUSTRY

- ✓ It is the largest industry of Pakistan.
- ✓ It provides employment opportunities to 50 % of industrial labor force.
- ✓ It contributes 7 % of G.D.P. to the economy of our country and also a source of foreign exchange.
- ✓ It has been established in different parts of our country i.e. Karachi, Faisalabad, Hyderabad, Multan, Lahore and other small cities like Jhang.
- ✓ After picking, the seed cotton is sent to ginning mills, here it is dried to reduce moisture and the dust etc is removed
- ✓ Then, the lint is separated from the seed by rotating saws, which don't allow the seeds to go through them. The lint is packed into bales, while the seeds are sent to Vegetable and Ghee industries
- ✓ Samples are taken from lint to assess its quality in terms of fibre length, consistency of length, colour etc; all of these factors determine the price of the bale
- ✓ The bales are sent to textile mills, where fibres from different bales are twisted and turned (spinning) together by computer controlled machines to achieve uniform thickness of fibre etc. This produces yarn.
- ✓ Then different yarns are interlaced to make cloth (weaving) which is also automated.
- ✓ The cloth is then cleaned, immersed in chemicals to enhance finishing and increase its price. Loose threads are also removed
- ✓ The cloth is then sent for dyeing, where a uniform colour scheme is applied to a length of cloth.
- ✓ After this it is sent for printing, where a series of design schemes are applied onto the cloth
- ✓ Sheets, pillow cases, shirts, bed lining etc all are made from this finished cotton cloth
- ✓ Cotton seeds, upon reaching the Vegetable and Ghee industry are washed and then passed through rollers. Thus, cotton seed oil is expelled leaving behind the cotton seed cake.
- ✓ The oil is used to make cooking oil or margarine etc
- ✓ The cake is used for dairy animal feed or even as fertilizer for crops
- ✓ Most of the cotton cloth is exported as it's sufficient to meet local demand.

- ✓ Cotton is only grown in some countries but it is used in many countries for many different products (yarn, cloth etc).
- ✓ Pakistani cotton is of good quality and sells at competitive prices (due to cheap labour employed during picking and processing in the mill etc)



Process A

- ✓ Rollers used to flatten cotton fibres

Process B

- ✓ The fibres are divided into threads;

Process C

- ✓ Threads is collected in moving cylinders where fast and slow rollers further divide slivers

IMPORTANT PROCESSES

Ginning:

- ✓ It is a process of separating cotton seed from lint for further processing in textile industries.

Spinning:

- ✓ It is the process of making yarns from the textile fibre is called spinning.

Weaving:

- ✓ It is a method of textile production in which two distinct sets of yarns or threads are interlaced at right angles to form a fabric or cloth.

Knitting:

- ✓ It is a method by which yarn is manipulated to create a textile or fabric

ADVANTAGES

- ✓ More jobs available, can reduce unemployment, higher income, formal employment, move from primary to secondary
- ✓ Beneficial to workforce, higher earnings.
- ✓ Will stimulate / increase trade, earn foreign exchange, improve balance of payments, pay off debts
- ✓ Stimulates construction of better roads, railways, power supply, water supply

DISADVANTAGES

- ✓ may cause less investment in other industries
- ✓ less land for food crops
- ✓ quality must be good, High competition from other countries
- ✓ leaf curl virus / other pests
- ✓ climatic limitations
- ✓ lack of skilled workforce
- ✓ may cause rural urban migration

Explain what can be done to improve Pakistan's cotton textile industry.

- ✓ Increase production of cotton crop / quality of cotton crop
- ✓ Research into solutions to leaf-curl virus
- ✓ Irrigation
- ✓ More investment/government support/tax holidays/loans
- ✓ Address child labour/environmental issues to ease international restrictions
- ✓ Modernise machinery/plant/technology
- ✓ Increase availability of power/electricity
- ✓ Training / education of workers

Explain why Lahore is an important centre of the textile industry.

- ✓ Cotton grown locally
- ✓ Water for washing
- ✓ Machinery from HMC/Taxila
- ✓ Labour supply - large skilled and unskilled population
- ✓ Power - national electricity grid, gas pipelines, oil pipeline, power stations
- ✓ Transport - good roads, railways transport north, south and west
- ✓ Tele-communications, access to internet,
- ✓ Education - skilled workforce, IT skills, foreign experts
- ✓ Dry port - to promote exports
- ✓ EPZ - to improve quality, better infrastructure
- ✓ Industrial estates - for accommodation, infrastructure
- ✓ Market - large population
- ✓ Investors/entrepreneurs - for capital
- ✓ Factories for all stages of production

Although cotton textiles is Pakistan's largest industry, it still faces challenges if it is to increase its global market share. Read the following two views about some of the challenges to the cotton textile industry in Pakistan:

A

Load shedding of electricity is a major challenge to the growth of the cotton textile industry.

B

Lack of skilled labour is a major challenge to the growth of the cotton textile industry.

Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider View A and View B in your answer.

I agree with view A more.

Why?

- ✓ Production stops;
- ✓ Orders cannot be completed;
- ✓ Workers may be sent home / not paid;
- ✓ Deadlines cannot be met;
- ✓ Lose market share if trading partners cannot rely on Pakistan.
- ✓ If buy generators so cost increases due to which difficulties faced in selling in international markets due to competitive prices.

SUGAR INDUSTRY

- ✓ Sugarcane is one of the cash crops in Pakistan.
- ✓ Pakistan's sugarcane production is 5th largest in the world
- ✓ Sugar is made from sugarcane. At the time of harvest, the cane is cut and tied in bundles and is quickly transported to sugar mills. This is so, as the sugarcane after being cut starts losing its sugar content and weight.
- ✓ At the plant, it is first weighed and the farmer is paid accordingly.
- ✓ Then the cane is washed and rocks etc are removed.
- ✓ Chalk is scrubbed to remove dirt and smell.
- ✓ Then it is passed through rollers to extract the juice.
- ✓ The fibrous thing left behind is known as bagasse
- ✓ Then lime is added into the juice to control pH, lime also stops decay of sucrose. Furthermore, lime also removes impurities by making them clump. The mixture is left like this for some time until impurities settle down at the bottom
- ✓ Sulphur Dioxide is passed through the juice, which bleaches many colour-forming impurities into colourless ones, so we get white sugar at end.
- ✓ The juice (excluding impurities) is heated to 70°C to evaporate water and increase sugar percentage. This process is repeated until colour changes from thin yellow to dark brown. This syrup is further concentrated under vacuum until it becomes supersaturated, and then seeded with crystalline sugar.
- ✓ On cooling, more sugar crystallizes from the syrup. This thick liquid is spun in a centrifuge to separate the white sugar crystals from the brown syrup. The syrup is returned to the boiling station and boiled again until there is no more sugar left to extract. The extracted sugar is dried and then packed before being shipped.
- ✓ The remaining product is called molasses. It has a low sugar content and further extraction of sugar from it is uneconomic
- ✓ Bagasse (fibrous material) is often used as a primary fuel source for sugar mills. It produces sufficient heat energy to supply all the needs of a typical sugar mill, with energy to spare which is sold. Bagasse is used to make disposable food containers, replacing materials such as Styrofoam, which is regarded as polluting. Bagasse is also used to make paper and is also made into cattle feed whereby it is mixed with molasses

- ✓ Molasses (black liquid) is also used for making yeast and spirits. It also relieves constipation etc

FERTILIZER INDUSTRY

- ✓ The fertilizer industry basically requires two materials, nitrogen and hydrogen.
- ✓ Phosphorous, Potash etc may also be required
- ✓ Nitrogen is obtained by liquid distillation of air; meanwhile hydrogen is produced by reacting methane with steam (water).
- ✓ For the purpose of methane supply, fertilizer factories are found close to gas pipelines in Pakistan and near sources of water. These factories are also present near markets, where the demand is high. The distances are small, so transport costs are low, and the demand can be met quickly
- ✓ Nitrogen and hydrogen are reacted to form ammonia, which is further reacted to form ammonium nitrate, urea, ammonium phosphate etc. Other minerals like phosphorous and calcium are added to have a balanced nutrient requirement for different crops
- ✓ Fertilizers are imported from Middle East to fulfil the majority of the demand for fertilizer in Pakistan.
- ✓ Most of the factories are located in Punjab to meet the demands of growing population there. Also, a gas pipeline network exists in Punjab.
- ✓ The use of HYV seeds has meant that to fulfil the increased nutrient requirement of these crops fertilizers must be applied.
- ✓ The flow of the Indus River in the Indus Plains has decreased a lot recently due to dam construction. So the river doesn't flood its banks much often and fresh alluvium is not laid much around its banks. Thus, fertilizers are required to fill the deficiency of minerals left in the soil, as they have been utilized by the previous crop.
- ✓ Desert soils are now being used for growing crops (Thal desert), where the sand dunes have been flattened and canals made. But these desert soils are very porous, so the minerals are quickly leached out of the topsoil. Thus more and more fertilizers are required
- ✓ In recent years, Pakistan has tried to reduce its fertilizers import so it could reduce its import bill. As a result, Pakistan will have more money to spend on things like education, health etc.
- ✓ Also, local production results in cheaper fertilizers as costs of transport are far lower and that employment is generated in local factories
- ✓ Following are the fertilizer companies: Engro fertilizer company, Fauji fertilizer company, Pak Arab fertilizer company, Fatima fertilizer company.

Process

- ✓ Rotten plants, animal waste and some required chemicals are put together in a big Iron container.
- ✓ The mixture is heated until it is converted into a thick paste. Now natural gas is added to it without burning in raw form.
- ✓ In last stage the paste is dried and its grains are made for easy application.

ADVANTAGES

- ✓ Higher yields
- ✓ More food production
- ✓ More agricultural exports, or improved balance of payments
- ✓ Reduced imports of fertiliser, or improved balance of payments
- ✓ Higher GNP
- ✓ Less debt
- ✓ Higher farm incomes / profits
- ✓ More jobs
- ✓ Cheaper cost of fertiliser
- ✓ More industrial goods (e.g. cotton)

BRICK INDUSTRY

- ✓ A kiln in which raw bricks are baked or burned.
- ✓ It is an important small-scale industry of Pakistan, it provides employment to people in rural areas where agriculture is not enough to sustain all the people.
- ✓ It is generally situated in desolate places away from the main cities and towns.
- ✓ There are 6000 (estimated) brick kiln in Pakistan, its share in GDP is approximately 1.5 %. In kilns the working days are hardly 240 - 260 in a year. On rainy days there is no work.
- ✓ To make bricks, raw clay along with 30% sand and water (amount varying due to choice) is put into steel moulds.
- ✓ They are compressed and then put into a brick kiln at around 900 degree celsius.
- ✓ The fuel is in Pakistan coal. Most of coal produced in Pakistan is used this way. A series of chemical reactions take place thus changing the chemical structure of the mud and makes it hard
- ✓ After being moulded the bricks are left to dry, and then piled into groups of ten (called tuttoos) and groups of twenty (called ghoris).
- ✓ Majority of bricks made are used locally for construction, canal lining etc
- ✓ They rely on intensive labour consists of males, women and children. They get a very small amount to make 1000 katcha bricks.
- ✓ In the preparation of 1000 katcha bricks, the whole family, i.e. women, children are involved but wages are given to a single person only.
- ✓ There are different kinds of workers at the kilns; Patheras: who make the unbaked bricks after mixing with water and clay. Bharaiwala: who load the unbaked bricks on donkeys and carry them to the kiln area. Jalaiwala: who feed coal into the furnace. Nakasiwala: who remove the baked bricks from the kiln.

Explain how bricks are made.

- ✓ Clay mixed with water
- ✓ Placed in moulds
- ✓ Dried (in sun)
- ✓ Baked (in kiln)

Effects on Environment

- ✓ The heavy smoke particles containing CO_2 , SO_2 , smoke, dust, nitrogen oxide and other harmful gases lead to air pollution and further leading to environmental hazards such as acid rain, global warming and ozone depletion.
- ✓ Acid rain and Ozone depletion results into health hazards and diseases such as skin cancer, eye diseases, Asthama, deafness (from noise).
- ✓ The trees are cut down in order to obtain wood required to burn bricks. This results in heavy deforestation and it makes the soil infertile.
- ✓ Agricultural land is reducing in the areas near brick kiln fields.
- ✓ The waste is being dumped into rivers and it has lead to water pollution with many marine species in danger.

Name types of air pollution that might be produced by a brickworks.

- ✓ Carbon dioxide/carbon monoxide
- ✓ Nitrogen oxides
- ✓ Sulphur dioxide
- ✓ Smoke
- ✓ Smell,
- ✓ Ash

Solution

- ✓ Measures to ask brick kiln owners to convert to natural gas from traditional methods, thus reducing large scale deforestation.
- ✓ The chimney of these Kilns should be high so that it reduces the degree of pollution, and reduces the vulnerability of harmful diseases such as asthma and Lung Cancer.
- ✓ People should use masks while working in Brick Kilns to avoid inhaling dangerous chemicals.
- ✓ Clean and efficient coal technologies should be used where coal is pretreated for complete combustion.
- ✓ Smoke should be filtered before it leaves the chimneys.

CEMENT INDUSTRY

- ✓ Raw materials for making cement are limestone, clay and sand.
- ✓ All of these after quarrying/mining are sent to the cement factories.
- ✓ These factories are located near to limestone sources as limestone is bulky and expensive to carry over long distances. This increases the per unit cost of production so profits are lowered. Also, accidents can occur as heavy trolleys are difficult to manoeuvre and they also damage the road infrastructure, so to minimize this damage, cement factories are located near limestone sources
- ✓ Here the limestone, clay and sand enter a crusher, where their size is reduced to size of a small rock.
- ✓ Then this mixture is analyzed in the labs and adjusted for proper percentage of different components.
- ✓ It is then grinded even finer into powdered form by wheel rollers
- ✓ It is then sent to pre-heater tower and then kiln.
- ✓ The kiln is a rotating drum which is attached to the pre-heating tower and slopes gently towards the ground.
- ✓ At the end of kiln we have a heat source such as coal or natural gas
- ✓ 1700°C temperatures change the powder into a new substance called clinker, which is in the form of pellets etc.
- ✓ The clinker is then broken down into cement powder.
- ✓ A small amount of gypsum is added to control the rate of setting of cement
- ✓ Majority of cement produced is for local consumption, although exports are gradually increasing.
- ✓ Cement is used to make factory walls, build walls of dams, line canals, build homes etc. Furthermore, structures made from cement and bricks are much more robust than structures made from wood etc.
- ✓ Cement is produced in large amounts in Pakistan because the raw materials required for cement making are found in Pakistan in abundance at a cheap price

EMPLOYMENT IN CEMENT INDUSTRY

- ✓ **Primary** - job related to mining raw materials (coal, limestone, gypsum) / mine / quarrymen;

- ✓ **Secondary** - job related to manufacturing product, packing / bagging product / factory worker;
- ✓ **Tertiary** - job related to lorry driving / sales / advertising / cleaning / maintenance.

Give reasons for the continuous increase in cement manufacture

- ✓ Industrial/economic development
- ✓ Urbanisation/construction
- ✓ Better/more housing, roads, offices, factories
- ✓ Higher living standards
- ✓ Population increase
- ✓ Raw materials readily/locally available

Explain the importance of concrete to the development of cities

- ✓ Developing / increasing population, industrialisation, offices, housing, ports, roads
- ✓ Better roads for transport
- ✓ Better bridges for access
- ✓ Stronger building materials
- ✓ Long-lasting / more durable / less chance of damage
- ✓ More modern appearance
- ✓ Improving slums / squatters / Kachi Abadi

IRON AND STEEL INDUSTRY

- ✓ The iron and steel industry requires 5 major components; coke (very pure carbon, which is a very processed form of coal), limestone, iron ore and fuel (mostly coal).
- ✓ Water (in huge amounts as an input) is used to cool the steel.
- ✓ Manganese (not Magnesium) and Chromite are used to make stainless steel
- ✓ Coke as well as iron ore are imported from countries like Australia, India and Brazil etc. Manganese is also imported
- ✓ Coke, limestone and iron ore are put into a blast furnace. Here the intense heat causes decomposition of the limestone. The carbon monoxide gas is produced which reduces iron ore into molten pig iron around 96% pure. After further purification of the iron, carbon and other metals are added to form different types of steel. This is done as iron is much reactive and softer than steel, so iron must be converted to steel
- ✓ When the steel solidifies, it is either sent in this form north to cities like Lahore etc. Here we have the Heavy Industry Complex at Taxila, which makes heavy machinery. In Punjab, we have a big and increasing population with increasing demand of steel for buildings, bridges etc
- ✓ Or at the steel mill it is cast into; Billets, Hot and Cold rolled coils/sheets, Galvanized rolled sheets or Slabs
- ✓ These products are at times exported to countries like China, UAE, Saudi Arabia, Romania, and Bangladesh.
- ✓ Although raw materials for making iron and steel are mostly imported

ADVANTAGES

- ✓ Cheaper supplies (than imports)
- ✓ Saves foreign currency / improves balance of payments / reduce imports
- ✓ Can be used for development / industrialisation e.g. construction, machinery
- ✓ Increases GNP / GDP
- ✓ Independence from other countries
- ✓ More work / employment opportunities

DISADVANTAGES

- ✓ Lack of raw materials
- ✓ High cost of imported iron / other raw materials

- ✓ Burden on economy / less development
- ✓ Lack of machinery / technology
- ✓ Lack of infrastructure / power supplies / roads etc.
- ✓ Lack of skilled labour
- ✓ Pollution
- ✓ May encourage rural-urban migration

METAL SHEETS/PLATES/SLABS



METAL ROLLERS/ROLLS



Why is over 50% of the output of Pakistan Steel sent north from Karachi to the Punjab?

- ✓ To Taxila
- ✓ Heavy engineering
- ✓ Machinery for industry and power generation/ construction / railways boilers etc.
- ✓ Construction of buildings/ bridges / pylons

It has been proposed that a large integrated iron and steel mill should be built on the outskirts of Rawalpindi, Punjab.

80 km² of land will be cleared for the mill and new infrastructure will be provided to the site by the local authorities. Large amounts of raw materials will be brought onto the site by road or rail and it is planned to produce up to 3 million tonnes of finished steel per year.

Evaluate how the new integrated iron and steel mill will affect the local people of Rawalpindi. Give reasons to support your answer. You should consider both benefits and problems in your answer.

Benefits

- ✓ Employment opportunities
- ✓ Improvements to roads/other infrastructure
- ✓ New settlements
- ✓ Educational/recreational facilities
- ✓ More trade for local shops
- ✓ More work for local related industries

Problems

- ✓ Farmland/grazing used for industries
- ✓ Air pollution/dust from chimneys, etc.
- ✓ Visual pollution/eyesore
- ✓ Land pollution
- ✓ Roads/railways congested
- ✓ Attracts migrants
- ✓ Uses large amounts of water/power

SPORTS GOOD INDUSTRY

- ✓ Sialkot is the major foci of sport goods production in the world. It accounts for around 40% of world football production accounting for some \$200 million exports annually
- ✓ Raw materials for making footballs are leather, stitches, yarn and glue.
- ✓ Furthermore, things like bats, hockeys etc all are made. These require wood, nails and polish/paint
- ✓ However, it must be remembered that due to the problem of child labour many European and American sports brands have stopped orders to Sialkot and Pakistan in general.
- ✓ The problem has been further compounded by the fact that due to loadshedding, many businesses have failed to meet their orders and thus their reputation has been tarnished. This has caused huge financial losses to the industry
- ✓ Even though most of sports goods production is exported due to modest local demand and high international demand

Describe how sports good industry operates in Pakistan?

- ✓ In small workshops / within household / small scale industry
- ✓ Uses manual labour / by hand / hand tools;
- ✓ Processes, e.g. cutting, sewing, stitching, shaping;
- ✓ The final stages are done in factories;
- ✓ Mostly work is conducted by unskilled / child labourers;
- ✓ Goods are manufactured to international standards;
- ✓ All stages are supervised for quality control.

ADVANTAGES

- ✓ Enhances traditional skills
- ✓ Uses local raw materials/saves import of raw materials
- ✓ Increases employment
- ✓ Work for women
- ✓ Increases family incomes/GDP
- ✓ More exports/trade

DISADVANTAGES

- ✓ Shortage of raw materials
- ✓ Cost of importing raw materials/machinery
- ✓ E.g. rubber/thread/leather
- ✓ Lack of skilled labour
- ✓ (Trade hindered by) child labour issues
- ✓ (Trade hindered by) quality issues

Why is a large proportion of the production of this industry exported?

- ✓ to make capital/improve trade balance/lessen debt/raises GDP
- ✓ to earn foreign exchange
- ✓ demand from abroad
- ✓ popularity of sports in the world
- ✓ competitive price/good quality
- ✓ good reputation of Pakistan
- ✓ less need in Pakistan/greater need abroad

How can Pakistan maintain and increase its exports of sports goods?

- ✓ improve quality/quality control
- ✓ more Export Processing Zones/EPZ
- ✓ more Dry Ports
- ✓ modernisation/machines to replace hand work
- ✓ training/skills
- ✓ innovation/new products
- ✓ ban child labour
- ✓ regular/reliable supply
- ✓ more factories/government incentives
- ✓ better roads/airports/telecommunications/uses of telecommunication
- ✓ advertising
- ✓ do deals with companies, e.g. Adidas

SURGICAL INDUSTRY

- ✓ Sialkot also is famous for its surgical instruments, whose export in 2008-2009 was around \$250 million.
- ✓ These instruments are made mainly in small workshops which employ many thousands of people.
- ✓ The quality of these instruments is strictly monitored
- ✓ However, it must be noted that in western societies there has been a growing concern about the use of child labour (unethical), exploited labour (low wage) and poor worker conditions (poor sanitary conditions, hot environment) in which these instruments are made.
- ✓ This has forced some firms to stop orders from contractors in Pakistan.
- ✓ The raw materials required for making surgical instruments are titanium, iron, chromium and nickel
- ✓ Most of production is exported due to modest local demand but demand from countries like UK and Germany etc is high

OIL REFINING INDUSTRY

- ✓ Pakistan has 5 major oil refineries.
- ✓ 2 are located at Karachi (Indus Refinery and Pakistan Refinery), 1 in Hab district of Balochistan (Khalifa Coastal Refinery), one in Attock (Attock Oil Refinery) and last but not the least the Mid-Country Oil refinery at Mehmood Kot
- ✓ Pakistan imports the majority of its oil requirements. They are refined at coastal refineries and sent inland for further processing etc
- ✓ Oil is refined by heating and boiling all of its components, which are then gradually cooled, the components which have a lower boiling point settle in the top halves of the boiling tower, while the heavier and higher boiling point components settle near the bottom,
- ✓ Petroleum gas like propane is used for combustion.
- ✓ Naphtha is used to make chemicals and plastics.
- ✓ Petrol is used by automobiles.
- ✓ Kerosene is used as jet fuel.
- ✓ Diesel oil is used as fuel by cars, trucks etc.
- ✓ Lubricating oils are used to make lubricants.
- ✓ Fuel oil is used as fuel by ships and asphalt is used to make roads.
- ✓ Most of crude oil is imported, while very low quantities of refined oil is imported

CRAFT INDUSTRY

- ✓ Craft industry are business models based on the production of handmade items.

Name an example of a craft industry

- ✓ Carpet
- ✓ Traditional textiles
- ✓ Embroidery
- ✓ Jewellery
- ✓ Ceramics
- ✓ Woodwork
- ✓ Metalwork
- ✓ Sports goods
- ✓ Pottery

In what ways is this type of industry important to the local economy?

- ✓ Employment
- ✓ Meet demand of local market
- ✓ Reduces rural-urban migration
- ✓ Uses local raw materials
- ✓ Uses waste materials
- ✓ Low investment in technology/cheap to set up
- ✓ Increases capital/earns money for the local community

TOURISM INDUSTRY

- ✓ Tourism is a phenomenon where by people move temporarily to places away from home, primarily for relaxation and tourism is a tertiary industry.

DOMESTIC TOURISM

- ✓ Domestic tourism is when people go on holiday within their own country.

FOREIGN TOURISM

- ✓ Foreign tourism is when people from a different country visit / go on holiday abroad.

WHY PEOPLE VISIT PAK?

- ✓ The UK has remained the main source of visitors.
- ✓ Other sources were India, Germany, Japan, Russia, China, Canada, Netherland, France, Sri Lanka, Australia, Saudi Arab, UAE, Italy, Norway, Denmark, Malaysia and Turkey.
- ✓ Most of the visitors from overseas visited main cities like Karachi, Rawalpindi, Islamabad and Lahore.
- ✓ This is mainly due to the nature of these visits, which are dominated by Pakistani's returning to visit friends and relatives.

For Business

- ✓ To attend trade delegations.
- ✓ For educational activities e.g. some educational institutions have hired foreign staff e.g. Principal and teachers.
- ✓ As staff of multinational companies such as oil companies, Foreign Banks e.g. Citibank, Standard Chartered Bank.
- ✓ As UNO officials who are working in a number of projects in Pakistan e.g. WHO (World Health Organization).

Visiting Families

- ✓ Many Pakistani's work abroad mainly in K.S.A, Kuwait, U.A.E, UK and U.S.A.
- ✓ They are not tourist but when on holidays, they come to visit their families, friends in Pakistan almost every year.

- ✓ They stay with family members and not in hotels.
- ✓ Many Students study abroad, they came here in their vacations to visit their families.

For Recreation

- ✓ Recreational purposes
- ✓ Historic Interests (Places)
- ✓ Museum
- ✓ Religious Faith
- ✓ Different culture to their own
- ✓ Festivals (Basant)
- ✓ Visiting Northern areas and beautiful lakes and gullies.

CULTURAL ATTRACTION IN PAKISTAN

- ✓ Pakistan is famous for its cultural heritage, which may attract a number of domestic and foreign tourists, the ones who are interested in History, culture and Archaeological research.

Archaeological Sites

- ✓ Moen-Jo-Daro
- ✓ Harappa
- ✓ Taxila.

Historic and Religious Sites

- ✓ Forts (Baltit), Lahore
- ✓ Mosques (Badshahi Lahore, Mahabat in Peshawar)
- ✓ Tombs (Chaukandi near Karachi, Makli near Thatta, Allam Iqbal and Ranjit Sings in Lahore)
- ✓ Sharines (Uch sharif)
- ✓ Khyber Pass
- ✓ Shalimar Garden

Modern Buildings

- ✓ Faisal Mosque,
- ✓ Parliament Building,

- ✓ President House
- ✓ Minar-e-Pakistan
- ✓ Jinnah Mausoleum.

Natural Attractions

- ✓ The northern areas of our country have many attractions which are the following.
- ✓ Mountains,
- ✓ Snowcapped peaks,
- ✓ Natural vegetations,
- ✓ Hiking,
- ✓ Snow falling,
- ✓ Steep slopes,
- ✓ River,
- ✓ Lakes and
- ✓ Glaciers,
- ✓ Beautiful valleys like Hunza, Skardu, Swat and Kaghan.

Explain why tourists visit valleys.

- ✓ Views of mountain peaks, snow, waterfalls, etc.
- ✓ Forests/orchards
- ✓ Fishing in River Kunhar
- ✓ Saiful Maluk Lake
- ✓ Jeep rides
- ✓ Walking/hiking/climbing/mountaineering/picnics
- ✓ Unique area/nothing like it anywhere else
- ✓ Cooler climate than plains

WHY PEOPLE VISIT FOLLOWING PLACES?

Islamabad

- ✓ Pakistan's capital, planned city and located at the base of the Margalla Hills.
- ✓ President House, Parliament and Faisal Mosque
- ✓ Rawal Lake and Dam
- ✓ Shakar Parrian
- ✓ Daman-e-Koh

- ✓ Pakistan Monument
- ✓ Jasmine Garden

Karachi

- ✓ Provincial Capital (Sindh)
- ✓ Sea Port
- ✓ International Air Port
- ✓ Clifton Beach
- ✓ Hawkes Bay and Paradise point
- ✓ The National Museum
- ✓ The Mausoleum of the Quaid-e-Azam

Lahore

- ✓ Provincial Capital (Punjab)
- ✓ Lahore Fort and Badshahi Mosuqe
- ✓ Data Darbar
- ✓ Jehangir's Tomb and Allama Iqbal Tomb
- ✓ Minar-e-Pakistan, Museum Shalimar Garden and other parks

Peshawar

- ✓ Provincial Capital (KPK)
- ✓ Khyber Pass
- ✓ Bala Hissar Fort
- ✓ Mahabat Khan Mosque
- ✓ Peshawar Museum.

Quetta

- ✓ Provincial Capital (Balochistan)
- ✓ Bolan and Khojak Pass
- ✓ Hill resort of Ziarat

ADVANTAGES

- ✓ Foreign exchange/currency
- ✓ Reduced debt
- ✓ Reduced trade deficit/improved balance of payments

- ✓ More jobs + development
- ✓ Higher incomes to locals
- ✓ Better local economy/demand for more services
- ✓ Better facilities, infrastructure, services, communications for locals
- ✓ Better cultural understanding
- ✓ Sustainable industry/will not run out like coal
- ✓ Reduces rural-urban migration
- ✓ Cultural/historic sites restored/maintained

Explain how local people can gain income from tourism in mountain areas.

- ✓ Making/sale of crafts
- ✓ Opening shops in village
- ✓ Guides (on tracks/trails)
- ✓ Named services e.g. hotels/restaurants
- ✓ Named transport services
- ✓ Offering accommodation in own home

Explain how tourism could help to develop some mountain areas. You may use examples in your answer.

- ✓ Infrastructure improves - roads / airports for travel/ electricity / water / gas / telecommunications
- ✓ Development of small scale industries, to raise living standards
- ✓ Money earned - for business people, shopkeepers, craftsmen, etc.
- ✓ Environmental improvement - e.g. re-forestation
- ✓ cultural change - meet other cultures/cultural exchange
- ✓ Less isolation - global awareness, trade
- ✓ Security improves
- ✓ Increased food production
- ✓ Improved health facilities - better sanitation, hospitals, healthy living
- ✓ Rural - urban migration reduced

PROBLEMS FACED BY TOURISM INDUSTRY

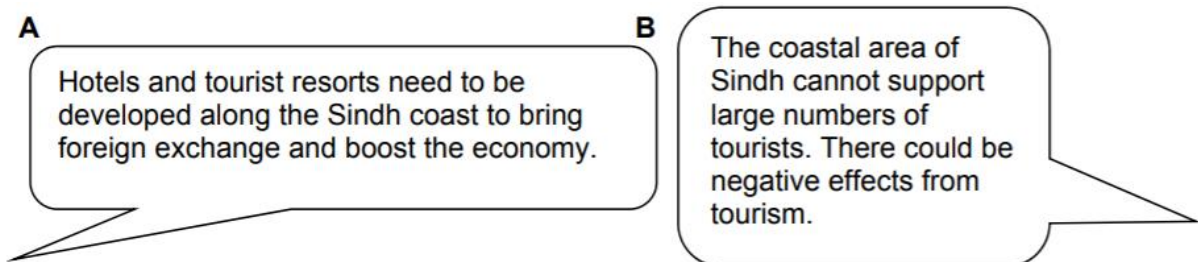
- ✓ Poor security/terrorism
- ✓ Bad management

- ✓ Lack of maintenance/cleanliness of tourist areas
- ✓ Poor transport infrastructure/poor road network/unpredictable railways/infrequent air access in north
- ✓ Lack of government funding/priority
- ✓ Little tourism infrastructure / few tour agencies / hotels not international standard

Why is capital important in the development of tourism?

- ✓ Building and maintenance of shops, hotels, etc.
- ✓ For Security
- ✓ Travel facilities/roads/jeeps/buses, etc.
- ✓ Provision of infrastructure
- ✓ Professional management
- ✓ Marketing and publicity/advertising

Read the following two views about the possibilities for tourism in Sindh province:



Which view do you agree with more? Give reasons to support your answer and refer to places or examples you have studied. [6]

I agree with view B more.

Why?

- ✓ Sensitive environment (threats to mangrove forests / fishing grounds)
- ✓ Tourists bring culturally unacceptable behaviour / dress code
- ✓ Tourists can pollute the environment with noise / litter / oil from jet skis, etc. (which disturbs local residents / is a danger to wildlife)
- ✓ Indus delta / most of Sindh coast unsuitable for development (swamps / marshes / creeks / forests)

- ✓ Karachi needs tourist industry infrastructure (e.g. no passenger ferry terminal)
- ✓ Declining / lack of tourist numbers
- ✓ Employment only seasonal
- ✓ Loss of livelihood due to construction of resorts (e.g. fishermen)

ATTRACTIONS



- ✓ beach for relaxing holidays / sunbathing / water sports / admiring scenic view of ocean from beach / beach cricket or similar activities / riding camels



- ✓ chairlifts (in Murree) for leisure and sightseeing / explore mountains / adventurous activities / photo opportunities / scenic viewing / greenery / admiring the scenic beauty / entertainment and thrills



- ✓ shopping / buy souvenirs / hotels / stay in resort / market / investigating local culture / eating traditional food



- ✓ experience mountains (in northern regions) / adventure tourism / mountain climbing / hiking / skiing / trekking / snowboarding



- ✓ historic buildings / investigate culture / study traditions / historical sites / spiritual visit

FORMAL SECTOR

- ✓ The people involved in this sector are employees of the government or a firm (not self-employed).
- ✓ They have regular pays and working hours are fixed.
- ✓ Also, the business is legally registered and contributes to the GDP.
- ✓ Goods of a specific quality are produced by skilled workers who operate machines.
- ✓ Quality of the goods is monitored and maintained.
- ✓ A proper work area (a building is used).
- ✓ Services like pension scheme/ health and insurance benefits may also be available

INFORMAL SECTOR

- ✓ The people are self-employed workers.
- ✓ Their monthly income is variable.
- ✓ Working hours aren't fixed.
- ✓ The business isn't registered with the government tax departments.
- ✓ Goods are sometimes not consistent in terms of quality (sometimes good and sometimes bad).
- ✓ These workers are usually unskilled (don't used machinery) and thus can work on streets and in their homes as well (they utilize their manual skills) etc.
- ✓ There are no pension schemes/ health and insurance benefits

To what extent can work in the informal sector improve the lives of those living in urban areas? Explain your answer. [6]

Possible

- ✓ Unemployable find work
- ✓ Source of income
- ✓ Can afford better diet / improved housing / education
- ✓ Suitable for unskilled / illiterate
- ✓ Can use traditional skills
- ✓ Goods / services available locally

Not Possible

- ✓ Poor working conditions
- ✓ Low / irregular incomes
- ✓ Too many people doing the same thing / too much competition
- ✓ Goods are poor quality

INDUSTRIAL ESTATES

- ✓ These are future centres of industry, which are located near major cities.
- ✓ The government actually buys the land and plans the estate.
- ✓ The government builds facilities like railways, roads, electricity, communications etc.
- ✓ A residential area for families of workers is also planned along with schools and hospitals etc.
- ✓ Access to a dry port may also be provided
- ✓ Cheap loans are also provided for people to invest.
- ✓ For this very purpose the government invites people to invest through advertisements etc.
- ✓ Investors then make 20-30% payment of plots etc (getting control of land in return).
- ✓ The government then starts construction of the industrial estate.
- ✓ This procedure can take around 2-4 years

Benefits

- ✓ Employment / jobs / raise incomes
- ✓ Goods for local needs
- ✓ Goods for export / more trade
- ✓ Increase GNP / GDP / increases national income / economic growth
- ✓ Reduce imports
- ✓ Attracts more investors / entrepreneurs
- ✓ Development of named infrastructure e.g. roads, power, telecomm
- ✓ Stimulates growth of service industries

Explain how the building of industrial estates could help to increase industrial production in Pakistan.

- ✓ Increase quality of goods
- ✓ Reliable power/telecomm supply
- ✓ Water supply/sanitation/cleanliness
- ✓ Roads, railways to and from the estate/transport network
- ✓ Attractive to investors/government incentives
- ✓ Opportunities for more technology/modernisation/specialisation

Problems

- ✓ Cost
- ✓ Lack of skilled labour
- ✓ Loss of agricultural land / trees
- ✓ Depletion of natural resources eg. water, gas
- ✓ Lack of infrastructure e.g. electricity, roads, water
- ✓ Lack of government support
- ✓ Pollution e.g. water, air, land
- ✓ Need for more imports e.g. machinery, raw materials, oil
- ✓ (Increases) rural-urban migration

List of Industrial Estates In Pakistan

- ✓ Karachi
- ✓ Lahore
- ✓ Gujranwala
- ✓ Jhelum
- ✓ Gujrat
- ✓ Hyderabad
- ✓ Peshawar
- ✓ Chakwal
- ✓ Daska etc

EXPORT PROCESSING ZONES

- ✓ These are places where goods are produced only for exports.
- ✓ There is no tax on imports and exports.
- ✓ Thus trade barriers like quotas and tariffs are all eliminated.
- ✓ High quality standard of goods are maintained (**essential**)
- ✓ The government builds the entire infrastructure like roads, railways, electricity, gas and water etc

Advantages

- ✓ Exports increase significantly; helping to pay off debts and reduce burden of loans etc
- ✓ Technology is brought to a country as foreign investors invest in the country. This means that local workforce is trained to use latest machines and learns new skills and techniques which increase production
- ✓ Employment is generated which raises the standard of living and reduces regional disparity
- ✓ For the workers it is also better as working conditions (pay, working hours) are good. This is required by international trading partners under labour laws etc
- ✓ Furthermore, the government can help in advertisement of the products, which are shown off during international trade exhibitions

List of EPZ In Pakistan

- ✓ Karachi
- ✓ Risalpur
- ✓ Sialkot
- ✓ Gujranwala